A389537

Residues of an alternating lacunary inverse

Dev Goyal

24 October 2026

AI disclosure. ChatGPT (OpenAI) generated the mathematical arguments, Lean proofs, and this manuscript in response to Dev Goyal's requests. Dev Goyal is the sole human author. The source sequence and conjecture or observation are credited to Paul D. Hanna. **Independent human review is pending.**

Abstract

The unique normalized integer solution of $A(A-x)=A^2$ has residue two modulo four exactly at indices $3\cdot 2^k$ above degree two. We identify it as a genuine compositional inverse and prove the complete pattern by exact integral halving followed by a characteristic-two recursion.

1 Source, notation, and scope

The source is Paul D. Hanna's OEIS entry A389537 [1], retrieved on 24 October 2026. Its initial sequence index is 1. We also retain the formal zero constant coefficient at index zero.

All series are formal. The set $\mathbb{N}$ includes zero, and $[x^n]U$ denotes a coefficient. A normalized series is $x+O(x^2)$; $O(x^r)$ means divisibility by x^r, over any coefficient ring used below. We write $U\circ V=U(V(x))$ only when $V(0)=0$. Compositional inversion, denoted $\mathrm{Rev}(U)$ or $U^{\langle -1\rangle}$, is distinct from multiplicative inversion of a unit. Congruences are coefficientwise; ordinary products and compositional iterates are distinct.

2 Formal algebra used in the proof

Lemma 2.1 (Coefficientwise construction)**.** *Let K be a commutative ring. Suppose an operator Φ preserves normalized series and, for $r\geq 2$, satisfies*

$$G-H=O(x^r)\quad\Longrightarrow\quad \Phi(G)-\Phi(H)=O(x^{r+1}).$$

Then Φ has exactly one normalized fixed point in $K[[x]]$. More generally, a residual equation whose degree-r condition determines the next unknown coefficient with multiplier one has a unique solution once its initial coefficients are fixed and its lower-degree conditions hold identically.

Proof. Begin with $G_0=x$ and set $G_{m+1}=\Phi(G_m)$. Normalization gives $G_1-G_0=O(x^2)$. Inductively, $G_{m+1}-G_m=O(x^{m+2})$, so each coefficient eventually stabilizes. Define G using those stable coefficients. The displayed implication ensures that each coefficient of $\Phi(G)$ equals the corresponding stable coefficient, hence $G=\Phi(G)$. If two fixed points agree below degree r, they agree through degree r; induction proves uniqueness.

For the residual version, set the next coefficient to the negative of the known residual term. Its multiplier is one, so the construction uses only ring operations. No earlier equation changes, and every equation is eventually imposed. The same coefficient comparison proves uniqueness. □

Lemma 2.2 (Units and reversion)**.** *Every constant-one series has a unique constant-one multiplicative inverse. Every normalized series has a unique normalized two-sided compositional inverse. These inversions, formal differentiation, and valid substitution commute with coefficient ring homomorphisms.*

Proof. For $UV = 1$ with $U(0) = 1$, the coefficient of degree $n > 0$ is v_n plus a finite expression in earlier coefficients of V, which determines v_n without division. For normalized G, the coefficient of degree $n > 1$ in $G(H)$ is h_n plus an expression in earlier coefficients of H, giving a unique normalized right inverse H.

To see the second inverse identity, construct a normalized right inverse J of H. Associativity gives $G = G \circ (H \circ J) = (G \circ H) \circ J = J$, and hence $H \circ G = x$. The same association proves uniqueness. Each coefficient of a valid composition is a finite polynomial in coefficients. Differentiation multiplies the degree-$(n+1)$ coefficient by $n+1$. Ring homomorphisms therefore preserve these operations and equations; uniqueness identifies the mapped inverses with the inverses of the mapped series. □

Lemma 2.3 (Binary support)**.** *There is a unique normalized integral series C satisfying $C = x + C^2$. Its reduction modulo two is*

$$P(x) = \sum_{k \geq 0} x^{2^k}.$$

More generally, for every integer $d \geq 1$, the unique zero-constant solution of $V + V^2 = x^d$ in $\mathbb{F}_2[[x]]$ is

$$V = P(x^d) = \sum_{k \geq 0} x^{d2^k}.$$

Proof. The operator $G \mapsto x + G^2$ satisfies Lemma 2.1, since $G^2 - H^2 = (G - H)(G + H)$ and $G + H = O(x)$. In characteristic two, every series V satisfies $V(x)^2 = V(x^2)$. This follows by truncating above any fixed degree and applying the polynomial Frobenius identity.

Thus $V + V^2 = x^d$ is a recursion in which the coefficient at an odd positive index is fixed immediately, while the coefficient at an even index n depends on the coefficient at $n/2 < n$. The constant coefficient is fixed to zero. This proves uniqueness. The proposed sum is well-defined coefficientwise because its exponents tend to infinity; adding its square cancels all terms except x^d. Set $d = 1$ to identify the reduction of C. □

Lemma 2.4 (Doubling and substitution)**.** *For $U, V \in (\mathbb{Z}/4\mathbb{Z})[[x]]$,*

$$2U = 2V \quad \Longleftrightarrow \quad U \equiv V \pmod 2.$$

If B, D have zero constant coefficient and $B \equiv D \pmod 2$, then $2H(B) = 2H(D)$ for every series H over $\mathbb{Z}/4\mathbb{Z}$.

Proof. For residues u, v modulo four, $2u = 2v$ exactly when their reductions modulo two agree. Apply this fact to each coefficient. Reduction modulo two commutes with substitution, giving the second assertion. In particular, this lemma never cancels two inside $\mathbb{Z}/4\mathbb{Z}$. □

3 A389537: an alternating lacunary inverse

3.1 Construction and the original equation

Over any commutative ring, define

$$R(x) = \sum_{k\geq 0} (-1)^k x^{2^k}. \tag{1}$$

This is a normalized formal series, and cancellation of successive terms gives $R(x) + R(x^2) = x$. It is the unique normalized solution of this equation: an odd degree above one has coefficient zero, and an even degree is determined by minus the coefficient at half that degree. Equivalently, $U \mapsto x - U(x^2)$ strictly increases the order of differences between normalized inputs, so Lemma 2.1 applies. In Lean the latter construction is used.

Let $A = R^{\langle -1\rangle}$ be the genuine normalized compositional inverse. Substituting A in $R + R(x^2) = x$ gives

$$x + R(A^2) = A.$$

Apply A to $R(A^2) = A - x$ to obtain the literal A389537 equation

$$A(A(x) - x) = A(x)^2. \tag{2}$$

Conversely, let normalized U satisfy (2), and let $V = U^{\langle -1\rangle}$. Applying V on the outside gives $U - x = V(U^2)$. Substituting V for x then gives $x - V = V(x^2)$. The uniqueness of (1) implies $V = R$ and hence $U = A$. We have therefore constructed the unique normalized integral solution of the original equation, not just an auxiliary series with a related congruence. Coefficient maps preserve this construction by Lemma 2.2.

3.2 Parity, exact halving, and the modulus-four result

Theorem 3.1. *For the preceding integer source series, $a_1 = a_2 = 1$, and for every $n > 2$,*

$$a_n \equiv \begin{cases} 2 \pmod 4, & n = 3 \cdot 2^k \text{ for some } k \in \mathbb{N}, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Modulo two, Frobenius changes the inverse equation to $R + R^2 = x$. Thus $(x + x^2) \circ R = x$. The uniqueness of normalized compositional inverses shows

$$A \equiv x + x^2 \pmod 2. \tag{3}$$

In particular, there is a well-defined integral series

$$H = \frac{A - x - x^2}{2} \in \mathbb{Z}[[x]], \qquad H(0) = 0. \tag{4}$$

This division takes place coefficientwise in the integers, after the divisibility proof (3); it is not division by two inside $\mathbb{Z}/4\mathbb{Z}$. The initial values $a_1 = a_2 = 1$ also follow directly from $R = x - x^2 + O(x^4)$ and $R(A) = x$.

Reduce (4) modulo four, retaining the notation H, and set $D = A - x = x^2 + 2H$. Both D and x^2 have zero constant coefficient, and they have the same reduction modulo two. Lemma 2.4 therefore gives $2H(D) = 2H(x^2)$. Expanding the source equation yields

$$\begin{aligned} A(D) &= D + D^2 + 2H(D) = x^2 + 2H + x^4 + 2H(x^2), \\ A^2 &= (x + x^2 + 2H)^2 = x^2 + 2x^3 + x^4. \end{aligned}$$

All equalities in this calculation are in $(\mathbb{Z}/4\mathbb{Z})[[x]]$. After subtraction,

$$2(H + H(x^2)) = 2x^3.$$

Applying the equivalence in Lemma 2.4, and writing $\overline{H}$ for reduction modulo two, gives

$$\overline{H} + \overline{H}(x^2) = x^3, \qquad \text{equivalently} \qquad \overline{H} + \overline{H}^2 = x^3.$$

Since $\overline{H}(0) = 0$, Lemma 2.3 identifies it uniquely:

$$\overline{H} = \sum_{k \geq 0} x^{3 \cdot 2^k}.$$

For $n > 2$, equation (4) gives $a_n \equiv 2[x^n]H \pmod 4$. The displayed support proves both residue cases. □

This is the modulus-four conjecture in A389537 [1]. The same entry's empirical observation about lengths of sign runs is a separate problem and is not a consequence claimed here.

4 Separate Lean proof and checked scope

The entry file is `lean/Entries/A389537.lean`. Its explicit theorem statements and proof terms use the namespace `FinalCollection.A389537`. The complete imported implementations and their shared mathematical dependencies are included in the collection. The following local theorem names give the proof map:

- `auxiliary_exists_unique`.
- `auxiliary_equation`.
- `inverse_right`.
- `inverse_left`.
- `normalized`.
- `source_equation`.
- `source_exists_unique`.
- `source_unique`.
- `series_mod_two`.
- `integer_decomposition`.
- `half_equation_mod_two`.
- `half_binary_support`.
- `residue_mod_four`.

The main implementation module is `Round3.LacunaryInverse`. The catalogue records the full original endpoint names and the new entry theorems. The project pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The collection report records source recompilation and transitive axiom checks. Accepted axioms are propositional extensionality, classical choice, and quotient soundness. All index conclusions above are unbounded; exact finite coefficient checks in the package provide separate corroboration.

Limits and status. The separate empirical sign-run observation is not proved. Integer halving precedes reduction modulo four. The manuscript consolidates completed results; it does not assert global novelty, exclusive priority, peer review, OEIS acceptance, or a new submission. Standard algebraic ingredients are not claimed as new discoveries.

References

[1] Paul D. Hanna, *OEIS A389537*, 3 November 2025. `https://oeis.org/A389537`. Retrieved 24 October 2026.